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Mechanically Verified Optimization and UPPAAL-Oriented Transformation of Timed Büchi Automata

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ABSTRACT Timed Büchi automata (TBAs) produced by translators or other tools can contain redundant timing constraints, resets, transitions, clocks, and locations, and may have disjunctive invariants that require transformation for UPPAAL-oriented use. We present a Rocq-verified pipeline that synthesizes and propagates timing bounds, removes contradictory transitions and dead resets, merges synchronously reset clocks and equivalent transitions and locations, and transforms the result to conjunctive guards and invariants. Every pass preserves the timed-word language of each input TBA under existential initial-clock semantics. The preservation theorems hold for every TBA represented by the formula-indexed clock interface, independently of whether the automaton was generated by the MTL translator. Optimization preserves language for every finite number of rounds; export preserves language and proves structural properties of the internal target representation. The functions are extracted to OCaml and used in `mtl2tba`; XML printing and its correspondence to UPPAAL are outside the Rocq theorem. Of 31 attempted upstream-generated configurations, 29 produced optimizer inputs. We also evaluate 16 translator-independent synthetic TBAs and four system configurations. Optimization reduces transitions on the completed inputs, while explicit export can increase edge counts. These benchmark-specific results do not establish a universal efficiency gain.

INDEX TERMS Rocq, formal verification, program extraction, timed automata, timed Büchi automata, UPPAAL, verified optimization.

I. INTRODUCTION

Timed Büchi automata (TBAs) are a common intermediate representation for real-time requirements and are also used as inputs to timed-language analysis tools. An automaton that is correct at the language level may still be inconvenient for downstream verification: its transition guards can repeat constraints, some resets may never affect a future guard, multiple clocks may evolve in lockstep, and the same behavior may be represented by many transitions. In addition, merging transitions can produce disjunctive guards and invariants, while UPPAAL-oriented automata are most useful with conjunctive guards and conjunctive upper-bound invariants.

An optimization pass is safe only when it preserves accepted timed words, including the treatment of clock resets, event labels, and Büchi acceptance. Several passes also interact. Propagating a lower bound may make a transition contradictory, but a synthetic nonnegativity guard can keep an otherwise dead clock live unless guards are normalized before reset analysis. Splitting a disjunctive invariant into copies also requires a condition on entry: choosing a copy whose invariant admits less time than another copy can change the accepted language.

This paper presents a mechanically verified post-processing and export pipeline for TBAs. The preservation theorems hold for every TBA represented by the formula-indexed clock interface, independently of whether the automaton was generated by the MTL translator. In this interface, clocks are named by the timed subformulas of a fixed formula; the formula only supplies clock names, and no proof of an optimization or export pass uses any other property of it. In Rocq (formerly Coq), we prove that each optimization and export operation preserves the automaton's timed-word language under the acceptance semantics formalized in the development. The optimizer theorem holds for every finite number of rounds, so the implementation can stop after reaching a structural fixed point or after its configured iteration bound without relying on convergence for correctness.

The transformations have two stages. The optimization stage synthesizes upper-bound location invariants, propagates target invariants backward into guards, propagates entry

lower bounds forward, removes contradictory and unreachable transitions, removes dead resets, merges clocks reset synchronously, tightens guards, and merges transitions and locations. The export stage merges transitions with equal source, target, label, and resets; eliminates disjunctive invariants by creating copies with conjunctive invariants and clock-difference entry conditions; splits disjunctive guards; normalizes the result; and emits symbolic transitions for drawing.

The verified transformations are extracted to OCaml and used by `mtl2tba`. Our earlier article [1] establishes the correctness of a specified $\mathrm{MTL}_{0,\infty}$-to-TBA translation. Its output may still contain redundant bounds, resets, clocks, transitions, and locations, and may not have the invariant shape needed for UPPAAL-oriented output. This paper begins with that TBA and verifies the later optimization and representation transformations. The preservation theorems quantify over every TBA in the formal input type, irrespective of provenance; that proof scope does not supply a verified parser for arbitrary third-party formats. Our evaluation includes both upstream-generated inputs and a separate translator-independent synthetic suite, and we keep those evidence sources distinct.

The contributions are:

- a Rocq development of language-preserving TBA transformations, including invariant synthesis and propagation, clock and state simplification, transition merging, and disjunctive-invariant elimination;
- a preservation theorem for every TBA of the formula-indexed clock interface and every number of optimization rounds, plus a verified export composition whose final guards and invariants are conjunctive;
- an extracted implementation and an explicit account of its interface and trust boundaries; and
- a reproducible evaluation on 29 completed optimizer inputs from 31 attempted upstream-generated configurations, 16 directly constructed TBA records designed to exercise the passes, and four system-model configurations, with the input-generation and measurement boundaries stated explicitly.

Our earlier article, "A Mechanically Verified Translation of $\mathrm{MTL}_{0,\infty}$ into Timed Büchi Automata" [1], establishes the source translation theorem, and this paper reuses its Rocq core and correctness files unchanged. The present paper starts from the resulting TBA and verifies the subsequent optimization and export transformations. The composition theorem in Section VI retains the backend-language contract as a hypothesis; neither the Spot executable nor the XML printer is covered by that theorem.

II. TIMED AUTOMATA AND THE PRESERVATION CONTRACT

A. TIMED WORDS AND TBAS

A timed word is an infinite sequence $w = (a_i, t_i)_{i\in\mathbb{N}}$, where a_i is an event from a finite alphabet Σ, timestamps satisfy $t_i \geq 0$ and are strictly increasing, and time diverges: $\lim_{i\to\infty} t_i = +\infty$. For each position i, let $\delta_i = t_{i+1} - t_i$ be the time from event i to the next event. A clock valuation maps a finite clock set X to $\mathbb{R}_{\geq 0}$. Guards are finite conjunctions of event literals and clock comparisons $x \bowtie c$ for $\bowtie \in \{<, \leq, =, \geq, >\}$; transitions also carry a set of clocks to reset.

A TBA is a tuple $\mathcal{A} = (L, \ell_0, X, E, Acc)$, where L is a finite location set, ℓ_0 is the initial location, X is the clock set, E is a finite set of transitions $(\ell, \alpha, g, Z, \ell')$, α is an event condition, g is a clock guard, $Z \subseteq X$ is a reset set, and $Acc \subseteq L$ is the set of Büchi-accepting locations. The starting automata in the verified optimization are represented without location invariants; synthesis adds them as a separate annotation. Exported invariants are conjunctions of upper bounds.

The Rocq development interprets transitions over an extended timed word that records clock values and reset decisions. Here v_i is the clock valuation at event i, and Z_i records the clocks reset by the transition on that event. Since a reset occurs at t_i, the next-event valuation includes the elapsed interval δ_i:

$$v_{i+1}(x) = \begin{cases} \delta_i, & x \in Z_i, \\ v_i(x) + \delta_i, & x \notin Z_i. \end{cases}$$

A run chooses one enabled transition per position and visits accepting locations infinitely often. The accepted language existentially quantifies over clock-consistent extensions of the base timed word, including the initial clock values. This is the acceptance notion of the preservation theorems; conventional initialization at zero is a separate deployment concern.

For an annotated automaton, an invariant must hold throughout the delay spent at its location. Since synthesized invariants are conjunctions of non-strict upper bounds and clocks are nondecreasing during a delay, holding at the event at the end of the delay is equivalent to holding throughout that stay. The proof formalizes this trace-model equivalence and uses it to show that the annotations and strengthening operations preserve timed words.

B. WHAT EXPORT MUST PRESERVE

The export must retain the same transition alphabet, Büchi marking, clock meaning, and timed-word language. It may change the number of locations, transitions, and clocks, and may introduce difference constraints in transition guards to choose among invariant copies. Such choices must not create or delete an accepted timed word. The target representation has conjunctive guards and conjunctive location invariants; a later symbolic representation groups transitions with identical source, target, and reset sets under Boolean labels.

The preservation theorems hold for every TBA represented by the formula-indexed clock interface, independently of whether the automaton was generated by the MTL translator. Their proof does not require properties of the MTL translation or Spot. An upstream correctness theorem can be composed with these results, but Spot's correctness and the soundness of the front end are outside the pass theorems.

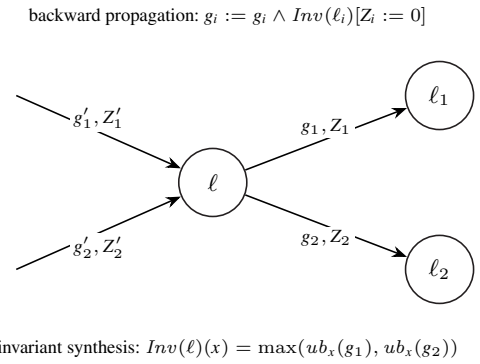


FIGURE 1. Invariant computation. Successor invariants strengthen outgoing guards; upper bounds of the resulting guards resynthesize the source invariant. Reset sets determine which target-clock constraints become constant.

III. VERIFIED OPTIMIZATION

After invariant synthesis, the optimization pipeline alternates timing propagation with simplification. Synthesized upper bounds expose how long an automaton can remain at a location. Propagating them to incoming guards can make lower and upper constraints jointly inconsistent; forward propagation supplies complementary lower-bound information. The pipeline then removes transitions, resets, clocks, and duplicate structure that no longer affect the language.

A. LOCATION INVARIANTS ON THE TRACE MODEL

A location invariant maps every clock x either to an upper bound $M \in \mathbb{R}$, standing for the constraint $x \leq M$, or to $+\infty$ (no constraint). We interpret automata with invariants on the extended words of Section II, as we do for automata without invariants. Along a run r, the automaton stays in location $r(i)$ between the events $i-1$ and i, and the invariant of $r(i)$ must hold during the whole stay, that is, at the clock values $\mathsf{val}_\rho(i,x) - \tau$ for every delay $\tau \in [0, \Delta_{i-1}]$, where $\Delta_{i-1} = t_i - t_{i-1}$ (and at the values $\mathsf{val}_\rho(0,x)$ for the initial location). This is the condition of Section II, evaluated at the event that ends the stay. Since clocks increase while time elapses, an invariant made of upper bounds holds during the stay if and only if it holds at the event that ends it (lemma `inv_during_iff_at_event`). For the two-sided invariants of Section III-C, a lower bound $x \geq m$ must likewise hold during the whole stay, hence at its beginning.

B. INVARIANT SYNTHESIS AND BACKWARD PROPAGATION

In the exported timed automaton, upper-bounded guards need invariants so that time does not elapse past all useful outgoing transitions. We compute location invariants from outgoing upper bounds and propagate successor requirements backward, using location invariants to constrain admissible time elapse as in standard UPPAAL models [2].

a: Upper bound of a guard.

For a conjunctive guard g and a clock x, let $ub_x(g)$ be the smallest bound m of the constraints $x \leq m$, $x < m$, and $x = m$ occurring in g, and $+\infty$ if g contains none. Other constraints are ignored. A valuation satisfying g satisfies $x \leq ub_x(g)$.

b: Invariant synthesis.

For a location ℓ with outgoing transitions $\ell \xrightarrow{g_i, Z_i} \ell_i$, $i = 1, \ldots, m$, the synthesized invariant is

$$Inv(\ell)(x) = \max_{1 \leq i \leq m} ub_x(g_i),$$

with $Inv(\ell)(x) = +\infty$ when ℓ has no outgoing transition. Thus a clock that some outgoing guard does not bound does not appear in the invariant. The bound is always non-strict, even when it comes from a strict guard $x < m$: the invariant only needs to be implied by the guard of the next transition, and strictness is kept in the guards themselves.

Theorem III.1 (Invariant synthesis)**.** *For every TBA $\mathcal{A}$, the automaton $\mathcal{A}$ with the synthesized invariants accepts the same timed words as $\mathcal{A}$.*

Proof. Adding invariants can only remove runs. Conversely, let r be an accepting run on ρ. At every event i, the transition taken from $r(i)$ is enabled, so its guard holds at the values $\mathsf{val}_\rho(i,x)$, which are therefore at most $Inv(r(i))(x)$. By the lemma `inv_during_iff_at_event`, the invariant of $r(i)$ holds during the whole stay. (Rocq: `add_invariants_correct`.) □

c: Backward propagation.

One propagation step strengthens every transition $\ell \xrightarrow{g,Z} \ell'$ with the invariant of its target after the resets:

$$g := g \wedge Inv(\ell')[Z := 0],$$

where $Inv(\ell')[Z := 0]$ replaces every bound $x \leq M$ on a reset clock $x \in Z$ by the constant constraint $0 \leq M$, that is, by $\top$ if $M \geq 0$ and by $\bot$ otherwise. Invariants are then synthesized again from the strengthened guards.

Theorem III.2 (Backward propagation)**.** *For every TBA $\mathcal{A}$ and every $n \in \mathbb{N}$, the automaton obtained by n propagation steps accepts the same timed words as $\mathcal{A}$.*

Proof. Strengthening guards can only remove runs. Conversely, consider an accepting run that takes $\ell \xrightarrow{g,Z} \ell'$ at event i and then a transition $\ell' \xrightarrow{g',Z'} \ell''$ at event $i+1$. The second guard holds at event $i+1$, so the values at that event are bounded by $Inv(\ell')$. Clock consistency gives $\mathsf{val}_\rho(i+1,x) = \Delta_i$ for $x \in Z$ and $\mathsf{val}_\rho(i+1,x) = \mathsf{val}_\rho(i,x) + \Delta_i \geq \mathsf{val}_\rho(i,x)$ otherwise; hence the strengthened guard $g \wedge Inv(\ell')[Z := 0]$ holds at event i. The argument uses clock consistency, so the theorem is stated on timed words. (Rocq: `propagate_accepts` and `propagate_n_accepts`.) □

Each round of the optimization pipeline of Section III-G applies one propagation step. The tool repeats the rounds until

TABLE 1. Verified post-processing passes. Each preservation theorem quantifies over every input TBA under the Rocq timed-word semantics. Detailed definitions and proof arguments follow in this section.

Pass	Rocq preservation theorem
Upper-bound invariant synthesis	`add_invariants_accepts`
Backward propagation (n steps)	`propagate_n_accepts`
Forward propagation of lower bounds	`forward_accepts`
Two-sided invariants	`add_two_sided_invariants_accepts`
Removal of contradictory transitions	`remove_contradictory_accepts`
Removal of outgoing transitions from locations with no incoming transition	`remove_unreachable_accepts`
Removal of dead resets	`remove_dead_resets_accepts`
Merging synchronously reset clocks	`merge_all_accepts`
Normalization of guards	`normalize_accepts`
Removal of unsatisfiable action labels	`remove_unsat_accepts`
Merging transitions	`merge_trans_accepts`
Merging equivalent locations	`merge_states_all_accepts`
Iterated pipeline (n rounds)	`optimize_accepts`
Merging transitions for export	`merge_transitions_accepts`
Maximal-delay invariant split	`split_accepts`, `split_inv_conjunctive`
All-copy invariant split	`split_all_accepts`, `split_all_inv_conjunctive`
Splitting of disjunctive guards	`explode_accepts`
Normalization of exported guards	`normalize_d_accepts`
Removal of copies never entered	`dprune_n_accepts`
Merging of the initial location	`dmerge_states_all_accepts`
Symbolic transitions	`symbolic_ext_accepts`
Complete export	`export_accepts`; shape: `export_guards_conjunctive`, `export_invariants_conjunctive`
All-copy export	`export_all_accepts`; shape: `export_all_guards_conjunctive`, `export_all_invariants_conjunctive`

the automaton is structurally equal to the previous one, as an OCaml value, or until 50 rounds have been performed; the bound guarantees termination, and the theorem holds for every number of steps, whether or not a fixed point is reached. On the 29 completed optimizer inputs, the greatest observed number of changed rounds is six (R6). The invariant computation at one location is summarized in the article's Figure 1.

d: Example.

We illustrate the algorithm on the automaton of Fig. 2, which contains a two-location cycle $\ell_1 \to \ell_2 \to \ell_1$ and a self-loop at ℓ_2. It has one clock x, the initial location ℓ_0, and the transitions

$$\ell_0 \xrightarrow{a,\ Z=\{x\}} \ell_1, \quad \ell_1 \xrightarrow{b,\ x\le 2,\ Z=\emptyset} \ell_2,$$
$$\ell_2 \xrightarrow{c,\ \top,\ Z=\emptyset} \ell_1, \quad \ell_2 \xrightarrow{d,\ x\le 1,\ Z=\emptyset} \ell_2.$$

The edge $\ell_1 \xrightarrow{b} \ell_2$ requires $x \le 2$, and the self-loop at ℓ_2 requires the tighter bound $x \le 1$.

1) *Synthesis.* $Inv(\ell_1)(x) = ub_x(x \le 2) = 2$. At ℓ_2, $Inv(\ell_2)(x) = \max(1, +\infty) = +\infty$, because the transition toward ℓ_1 does not bound x. At ℓ_0, the only outgoing guard is $\top$, so $Inv(\ell_0)(x) = +\infty$.
2) *Propagation.* The transition $\ell_2 \xrightarrow{c,\ Z=\emptyset} \ell_1$ becomes guarded by $\top \wedge Inv(\ell_1) = (x \le 2)$. The transition $\ell_0 \xrightarrow{a,\ Z=\{x\}} \ell_1$ receives $(0 \le 2) \equiv \top$ and is unchanged.

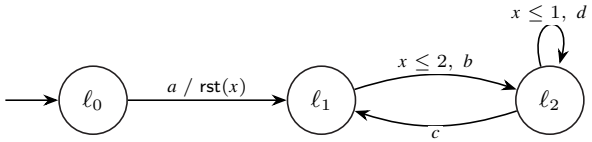


FIGURE 2. Automaton used to illustrate invariant propagation, with one clock x and actions a–d. Edges are labeled "guard, action / resets" (the guard is omitted when it is $\top$); only the edge $\ell_0 \to \ell_1$ resets x. The incoming arrow marks the initial location; acceptance plays no role here. The computed fixed point is $Inv(\ell_1) = Inv(\ell_2) = (x \le 2)$.

3) *Synthesis.* At ℓ_2, the outgoing guards are now $x \le 1$ and $x \le 2$, so $Inv(\ell_2)(x) = \max(1, 2) = 2$.
4) *Propagation.* The transition $\ell_1 \xrightarrow{b,\ x\le 2} \ell_2$ receives $x \le 2$, which it already contains, and the self-loop at ℓ_2 receives $x \le 2$, which is implied by $x \le 1$. Nothing changes.

The fixed point is $Inv(\ell_0) = \top$ and $Inv(\ell_1) = Inv(\ell_2) = (x \le 2)$. While the automaton remains in the cycle $\ell_1 \leftrightarrow \ell_2$, the clock x cannot exceed 2, and at every reachable valuation at least one outgoing transition of the current location is enabled. The invariants do not remove every blocking behavior: since x is never reset in the cycle, time is blocked once $x = 2$. They make the upper bounds of the property automaton explicit, and Theorems III.1 and III.2 guarantee that they do not change its language.

C. FORWARD PROPAGATION OF LOWER BOUNDS AND SIMPLIFICATION

a: Entry bounds.

Dually, for a transition $t = \ell \xrightarrow{g,Z} \ell'$ and a clock x, the *entry bound* of x along t is 0 if $x \in Z$ and otherwise the largest bound $lb_x(g)$ of the constraints $x \geq m$, $x > m$, and $x = m$ of g, if any. The entry bound of x at a location ℓ' is the minimum of the entry bounds of x along the transitions entering ℓ'; for the initial location, the initial entry, with bound 0, also counts. If some entering transition gives no bound on x, the clock x has no entry bound at ℓ'. Since clocks are nonnegative and do not decrease while time elapses, every clock with entry bound m at ℓ' satisfies $x \geq m$ during every stay in ℓ'.

b: Forward propagation.

The forward step adds $x \geq m$ to the guard of every transition leaving ℓ', for every clock x with entry bound m at ℓ' (Rocq: `forward_accepts`). The entry bounds can also be read as lower-bound invariants: the two-sided automaton that carries the synthesized upper-bound invariants and the entry bounds as lower-bound invariants accepts the same timed words (`add_two_sided_invariants_accepts`). UPPAAL rejects lower-bound invariants, and their only role is to make infeasible transitions visible: after forward and backward propagation, a transition whose source location bounds a clock from below and whose target bounds it from above carries both bounds in its guard.

c: Simplification.

Three simplifications follow. First, a transition whose guard contains contradictory constraints is removed: a lower bound and an upper bound on the same clock with $u < l$, or with $u = l$ and one of them strict, or an upper bound $x \leq u$ with $u < 0$ or $x < u$ with $u \leq 0$. Such a transition is never enabled (Rocq: `remove_contradictory_accepts`). Second, the transitions leaving a non-initial location that has no entering transition are removed: such a location never occurs on a run after position 0 (Rocq: `remove_unreachable_accepts`). The iteration of the pipeline removes chains of such locations one step at a time. A location whose only entering transitions are self-loops keeps them, but it is unreachable from the initial location; the tool counts and prints only the locations reachable from the initial one and their transitions. Third, a transition whose action label is unsatisfiable, because it requires two different actions or an action and its negation, is removed (Rocq: `remove_unsat_accepts`). Such labels arise when the translator receives an action a and the atom $\neg a$ of $\mathcal{T}(f)$ as independent propositions (the tool option `-indep-neg` of the translation article; its default negates the atom for a, which avoids most of them) or when a cube requires two different actions; removing them also enables clock merging, since the resets of a transition that can never be taken would otherwise prevent it.

D. DEAD RESETS AND MERGING OF SYNCHRONOUSLY RESET CLOCKS

a: Dead resets.

A clock x is *live* at a location ℓ if some transition that tests x in its guard can be taken from ℓ, possibly after other transitions. The live locations are computed as a backward closure, along all transitions, from the sources of the transitions that test x; the other locations are *dead* for x. Since the correctness of the pass does not depend on how this closure is computed, the pass checks the two properties that the proof uses, namely that no transition leads from a dead location to a live one and that none tests x from a dead location, and then removes x from the reset sets of the transitions that enter a dead location. After such a transition, the value of x is never read again, so whether it is reset does not matter. The proof builds, from every accepted extended word, an extended word that differs only in the reset decisions of x, and conversely (Rocq: `remove_dead_resets_clock_accepts`). The pass is applied to every clock (`remove_dead_resets_accepts`).

b: Synchronous clocks.

The verified translation shares the clock of syntactically identical timed subformulas. Clocks of *different* subformulas can also be merged when they are reset synchronously. Two clocks $x \neq y$ are *mergeable* if

(i) every transition resets x if and only if it resets y;
(ii) no transition enters the initial location; and
(iii) every transition leaving the initial location resets x and does not test it.

Then, after the first event, x and y always have the same value, and every test of x can be replaced by the same test on y (Rocq: `merge_accepts`). The condition is checked by a Boolean function proved to imply it, and the pass tries all pairs of reset clocks (`merge_all_accepts`). The merged clock x is no longer tested, so its remaining resets are removed by the dead-reset pass in the next round. This verified merging is restricted to synchronously reset clocks; merging clocks whose values coincide for other reasons is not covered.

E. NORMALIZATION OF GUARDS

Backward and forward propagation add constraints to guards. Every time a guard is strengthened, it is *tightened*: a constraint implied by another constraint of the same guard on the same clock is dropped, so that a guard keeps at most one lower bound and one upper bound per clock (an equality counting as both); for instance, $x \leq 5 \wedge x < 2$ becomes $x < 2$. The implication test compares bounds and strictness, and is proved sound (Rocq: `tighten_holds`). Normalization tightens every guard and also removes the lower bounds $x \geq m$ with $m \leq 0$ and $x > m$ with $m < 0$, which every clock value satisfies; it is needed after the merging of clocks, which renames constraints. The removal of trivial lower bounds relies on the nonnegativity of clock values, so the theorem is stated on timed words (Rocq: `normalize_accepts`).

F. MERGING OF TRANSITIONS AND OF LOCATIONS

a: Transitions.

Two transitions with the same source, target, and resets are merged when the union of their enabling conditions is again a conjunction. A transition is removed when another transition with the same source, target, and resets is *weaker*, that is, when its label and guard imply those of the other transition, constraint by constraint (subsumption). Two transitions whose labels, or guards, differ only by a literal and its negation, such as p and $\neg p$, $x < 2$ and $x \geq 2$, or $x \leq 5$ and $x > 5$, are replaced by one transition without that literal (resolution). The pass preserves the language because every enabled transition has an enabled counterpart with the same source and target, and conversely (Rocq: `merge_trans_accepts`). The other unions are disjunctive; they are formed by the export (Section IV-B).

b: Locations.

A location ℓ' is merged into a location ℓ when both are accepting or both are not, and when (i) every transition leaving ℓ' is subsumed by a transition leaving ℓ, and (ii) every transition leaving ℓ is subsumed by one leaving ℓ'. A transition u subsumes t when it has a weaker label, the same resets, the same target once ℓ' is replaced by ℓ, and every constraint of its guard is implied by a constraint of the guard of t. In (i) the guard of t is conjoined with the *entry invariant* of ℓ': the lower bounds $x \geq m$ that hold at every visit of ℓ'. They are the entry bounds of Section III-C, except that a self-loop of ℓ' that does not reset x is ignored, because it preserves a lower bound (Rocq: `entry_lb_loop_sound`). Both orders of a pair are tried. The transitions entering ℓ' are redirected to ℓ, and those leaving ℓ' are removed. A run of the original automaton becomes a run of the merged one by replacing ℓ' by ℓ, using (i) and the invariant; conversely, a run of the merged automaton is followed in the original one, taking in ℓ' a transition that subsumes the one taken in ℓ, by (ii) (Rocq: `merge_states_all_accepts`; "state" in a Rocq name means location). The condition on the Büchi marking is necessary: merging an accepting location with a non-accepting one would accept runs that visit only the latter infinitely often.

G. ITERATED OPTIMIZATION

In right-to-left function-composition order, one optimization round is

$$\begin{aligned}\text{optimize_step} = \text{merge_states}\\ \circ\ \text{merge_trans} \circ \text{normalize}\\ \circ\ \text{merge_clocks} \circ \text{dead_resets}\\ \circ\ \text{normalize} \circ \text{unreachable}\\ \circ\ \text{unsat_labels} \circ \text{contradictory}\\ \circ\ \text{forward} \circ \text{propagate}.\end{aligned}$$

The order is deliberate. Forward propagation can add $x \geq 0$ after a reset; the first normalization removes this tautology before dead-reset analysis, which would otherwise treat the clock as live. Each operation may enable later simplifications, so the driver repeats the round until the TBA is structurally unchanged or 50 rounds have run. The theorem is independent of this stopping rule and holds for every n.

Theorem III.3 (Optimization)**.** *For every TBA $\mathcal{A}$ and every $n \in \mathbb{N}$,* $\mathsf{optimize}(n, \mathcal{A})$ *accepts exactly the timed words accepted by $\mathcal{A}$.*

Proof. Every constituent operation preserves acceptance, as listed in Table 1. Their composition preserves acceptance, and induction on n proves the statement for every number of rounds. The mechanized result is `optimize_accepts`. □

The extracted function also returns synthesized upper-bound invariants and entry bounds. The tool compares successive automata structurally and stops when they are equal or after 50 rounds. The theorem holds independently of whether iteration reaches a fixed point or uses all 50 rounds. Among the 29 completed benchmark inputs, the largest observed number of changed rounds is six (R6); this is a measured property of these inputs, not a termination assumption in the proof. The practical 50-round cap is a driver policy, not a premise of `optimize_accepts`.

IV. MECHANICALLY VERIFIED ELIMINATION OF DISJUNCTIVE INVARIANTS

A. A WORKED EXAMPLE: SELECTING A LONGEST ADMISSIBLE DELAY

Suppose the invariant synthesized at a location q is $I_x \vee I_y$, where $I_x \equiv x \leq 3$ and $I_y \equiv y \leq 4$. From the entry valuation $(x, y) = (0, 0)$, the original disjunction allows a stay of up to four time units: I_y remains true after I_x has expired. A single copy carrying only I_x would stop the run after three units and could lose a timed word whose next event occurs at time four. The copies therefore cannot be selected by an arbitrary disjunct.

The split creates q_x with invariant $x \leq 3$ and q_y with invariant $y \leq 4$. A copy is enabled on entry only when it admits a delay at least as long as every other copy. With the difference-constraint convention $u - v \leq c$, the entry predicates are

$$G_x \equiv x \leq 3 \ \wedge\ x - y \leq -1, \qquad G_y \equiv y \leq 4 \ \wedge\ y - x \leq 1.$$

At $(0, 0)$, G_y holds and G_x does not, so the four-unit branch is selected. At the tie valuation $(0, 1)$, both predicates hold and both copies admit the same maximal delay. Comparing residual times $3 - x$ and $4 - y$ requires the clock-difference constraints; separate unary guards cannot express their ordering.

The reset-adjustment step is visible when an incoming edge resets x but leaves y unchanged. Write y^- for the value of y just before the edge; immediately after the edge, $x = 0$ and $y = y^-$. Substituting this reset into the two entry predicates gives

$$G_x[x := 0] \equiv y \geq 1, \qquad G_y[x := 0] \equiv y \leq 1.$$

Thus $y^- > 1$ admits the x-copy, $y^- < 1$ admits the y-copy, and the tie admits both. The comparison of two clocks has become a unary comparison because one clock was reset. If the edge resets both clocks, a difference constraint becomes a constant truth test; if it resets neither, the diagonal comparison remains. These are exactly the three reset cases handled by the verified guard-substitution function.

B. GENERAL SPLIT CONSTRUCTION AND PRESERVATION

For a general disjunction $I_1 \vee \cdots \vee I_r$, the construction makes one copy per irredundant conjunction U_k of non-strict upper bounds. At entry valuation v, the largest delay admitted by U_k is

$$d_k(v) = \min_{(x,b)\in U_k} (b - v(x)), \qquad d_k(v) = +\infty \text{ when } U_k = \varnothing.$$

The entry predicate for copy k is

$$G_k(v) \equiv U_k(v) \ \wedge \ \bigwedge_{\ell=1}^{r} d_\ell(v) \le d_k(v).$$

Thus a copy is admitted only if its invariant holds at entry and no other copy permits a strictly longer stay. To encode the comparison without introducing a new clock, the implementation chooses an upper bound $y \le b_y$ in U_ℓ and requires it to be reached no later than every bound $x \le b_x$ in U_k. This gives $v(x) - v(y) \le b_x - b_y$. The disjunction over possible minimizing bounds is represented in DNF. An incoming edge reset set Z is handled by substituting zero for clocks in Z before conjoining G_k with the edge guard: a difference constraint with two reset clocks becomes a constant test, one reset clock gives a unary comparison, and no reset clock leaves a diagonal constraint.

The copy invariants are generated from the outgoing transition guards. If $\mathrm{ub}(c)$ extracts the non-strict unary upper bounds from a conjunctive guard clause c, the coverage contract is

$$\forall t \in \mathrm{Out}(q),\ \forall c \in \mathrm{DNF}(g_t),$$
$$\exists U \in \mathcal{U}(q) : \quad \mathrm{ub}(c) \Rightarrow U.$$

Here t has source q and guard g_t; implication means that every valuation satisfying all extracted upper bounds also satisfies every bound in U. The Rocq lemma `disj_inv_cover` proves this coverage for the deduplicated list $\mathcal{U}(q)$; it is an internal construction invariant, not a precondition imposed on callers. At the endpoint of any taken transition, the selected guard clause therefore entails a generated upper-bound conjunction. Since clocks only increase during the stay, those non-strict upper bounds also hold at each earlier valuation. This coverage argument is the key completeness condition for both the verified split and the all-copy comparison below.

All copies inherit the original location's accepting mark. No silent edge is added, and the proof relates runs by projecting each copy back to its original location. Thus repeated visits to an accepting location correspond to repeated visits to accepting copies, preserving Büchi acceptance. In Rocq, `split_accepts` proves timed-word preservation when the input invariant function is everywhere absent; `of_tba` establishes that precondition for the TBA export pipeline. The split invariant is absent or one conjunction per location by `split_inv_conjunctive`; the full export shape is established by `export_guards_conjunctive` and `export_invariants_conjunctive`. These are the internal representation theorems; the XML printer is not verified.

The optimized automaton has normalized conjunctive guards. Export first merges transitions that agree on source, target, event label, and resets, placing their guards into a disjunction. The location invariant is then represented by disjunctions of conjunctions of upper bounds. UPPAAL-oriented output requires each such invariant to be conjunctive because its location syntax admits a conjunction of clock conditions, so a disjunction must be encoded by a structural transformation [2]. Retaining a disjunction directly would not meet the target representation.

Every incoming edge is replicated to each regular target copy and conjoined with its reset-adjusted entry predicate; outgoing edges are replicated from every copy. This replication is essential: copying only the location and assigning incoming edges to one arbitrarily chosen disjunct can truncate a stay and lose a timed word. The initial location uses a dedicated invariant-free copy because the first timed-word event has no incoming edge.

The representation cost follows directly from this construction. Let $K = \mathsf{kp}(D)$ be the number of source-copy slots used by the implementation and let $r(q)$ be the number of regular invariant copies of location q. Before guard explosion, each original edge t contributes $K\, r(tgt(t))$ split transition records. After guard explosion, the corresponding explicit-edge count is

$$E_{\mathrm{split}} = \sum_{t\in T} K \sum_{k<r(tgt(t))} \left|\mathrm{DNF}(g_t \wedge \mathrm{subst}_{Z_t}(G_k))\right|.$$

This count is taken before later normalization and pruning. It exposes two sources of growth: the number of invariant copies at the target and the DNF size of reset-adjusted entry guards. It is a structural count of the construction, not a complexity bound for the entire exporter.

Language preservation does not require selecting a copy with the longest admissible residence interval. We therefore mechanized a second policy, called the *all-copy split*: it retains the same conjunctive invariant copies, invariant-free initial copy, and accepting marks, but sends every incoming transition to every regular target copy with its original guard and reset set. It has no reset-adjusted entry predicate. The copy coverage property `disj_inv_cover` is established inside the Rocq construction, not assumed from an external parser.

Theorem IV.1 (All-copy split preservation)**.** *Let D be an input DTA with no pre-existing location invariants. The all-copy construction computes its upper-bound disjunctions and proves `disj_inv_cover` internally. Then the all-copy variant and D accept the same timed words.*

Proof. For soundness, project a run of the split automaton from each copy to its original location. Each copied transition keeps its original label, guard, and resets. The regular copies add their associated upper-bound conjuncts, but the input automaton has no location invariants, so projection forgets only this additional condition. The initial copy has no invariant, as required by the input contract. Accepting marks are inherited, so infinitely many accepting visits are preserved.

For completeness, fix an accepting run of D. At each non-initial location occurrence, consider the outgoing transition used by that run and a DNF clause c of its guard that holds at the event ending the stay. By `disj_inv_cover`, some copy invariant U_k is implied by the upper-bound part of c. Hence U_k holds at that ending event. Each atom of U_k is a non-strict upper bound, so it also holds at every earlier valuation in the stay: clocks only increase during delay, and resets occur at its beginning. Thus copy k satisfies its invariant throughout this stay. The preceding transition is available to every regular target copy, and every outgoing transition is available from each source copy. Choosing copy k on entry therefore lifts this occurrence of the run. The initial occurrence uses the dedicated invariant-free copy. Making this choice at each occurrence preserves the event sequence and resets; inherited accepting marks give infinitely many accepting visits in the lifted run. The two inclusions establish language equality. □

The maximal-delay policy establishes a stronger local copy-choice property: at a given entry valuation, every admitted copy has an admissible residence duration at least as long as that of any other copy. The helper lemma `best_spec` proves maximization, including $+\infty$ for an empty conjunction; ties remain admissible. This is neither global deadlock freedom nor a theorem about time divergence. Compared with all-copy replication, maximal-delay entry selection can introduce clock-difference constraints and more DNF clauses, while it may avoid entering copies that expire strictly earlier. The measured alternatives exhibit a trade-off (Table 2): on a two-clock example all-copy uses four more explicit edges but fewer guard atoms and no diagonal constraints; on a six-branch family both produce 68 edges, while all-copy uses fewer guard atoms and no diagonal constraints. All-copy is faster on the small case and slower on the six-branch case in these measurements. These are synthetic measurements, not a general performance ordering. The default remains the existing maximal-delay transformation; all-copy is separately verified and exposed for comparison.

V. VERIFIED UPPAAL-ORIENTED EXPORT

A. GENERATED CONTINUATION EXAMPLE

Figure 3 follows an actual current tool-generated automaton from the reset-completion boundary through optimization and export. It does not repeat the upstream MTL translation. In this instance, two optimization rounds remove nine transitions; export adds three explicit transitions while producing the required conjunctive internal shape.

The export continues by splitting disjunctive guards into conjunctive transitions, normalizing, pruning unreachable copies, and merging the initial location when an equivalent copy exists. A final symbolic pass groups transitions with the same source, target, and resets and stores their labels as Boolean conditions for drawings.

B. WHOLE-EXPORT PRESERVATION AND OUTPUT SHAPE

Theorem V.1 (Export preservation)**.** *For every invariant-free TBA provided to the verified export interface, the explicit export preserves its timed-word language. The resulting automaton has conjunctive transition guards and conjunctive location invariants.*

Proof. The explicit construction composes verified transition merging, invariant splitting, guard splitting, normalization, pruning, and initial-location merging; its timed-word preservation result is `export_accepts`. The output shape is established by `export_guards_conjunctive` and `export_invariants_conjunctive`. A separate symbolic grouping pass preserves timed-word acceptance by `symbolic_accepts` (with the extended-word result `symbolic_ext_accepts`). □

For reuse with another producer or checker, the theorem's output contract is summarized in Table 3. The theorem holds for every TBA of the formula-indexed clock interface: it requires no MTL provenance, determinism, or particular upstream producer.

The saved generated files for the R2 input, optimized automaton, and export are provided in `evaluation/generated\_examples/R2/`. Their run metadata and the distinction between this current snapshot and the preserved five-run medians are recorded alongside the files. This example shows an actual optimizer/export interaction, while the separate synthetic ablation in Section VIII isolates pass effects on direct TBA records.

VI. COMPOSITION WITH THE MTL-TO-TBA CORRECTNESS THEOREM

The earlier article [1] proves that its specified compiler preserves the timed-word semantics of every applicable formula, provided the propositional Büchi backend satisfies its stated language contract. The results of this paper then preserve the language of the resulting TBA through any finite number of optimization rounds and through export. The repository already proves the composition as `MTL_to_exported_correct_with`; its statement is

Theorem VI.1 (End-to-end composition)**.** *For every $n \in \mathbb{N}$, formula f, backend automaton A and timed word w,*

$$\Big(\forall s : \mathsf{pword}(f),\ \mathsf{action_consistent}(s) \Rightarrow \mathsf{PBA_accepts}(A, s) \leftrightarrow \mathsf{psat}($$

$$\Longrightarrow \quad \mathsf{msat}(w, 0, f) \ \leftrightarrow\ \mathsf{DTA_accepts}(D_$$

$$D_n = \mathsf{export}($$

$$\mathsf{optimize}(n, \mathsf{compile_with}(A)))$$

TABLE 2. Measured split policies on direct synthetic TBAs. Times are means over 200 in-memory export calls.

Input	Policy	Reach. loc.	Edges	Guard atoms	Diagonal atoms	Time (ms)
Two-clock	Maximal delay	3	6	12	6	0.831
Two-clock	All-copy	3	10	8	0	0.712
Six-branch	Maximal delay	5	68	320	204	15.990
Six-branch	All-copy	5	68	64	0	28.731

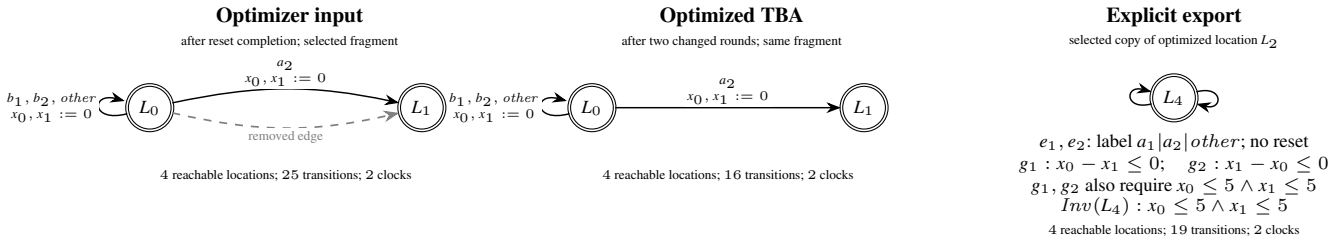


FIGURE 3. Selected subgraphs from the actual generated R2 files. The dashed input edge is absent after optimization; only one of the nine removed transitions is shown. The first two panels show a retained edge and loop. The export panel follows the optimized L_2 loop to its L_4 copy, where one source loop becomes two conjunctive-guard loops. The annotations list their diagonal atoms, shared unary bounds, and invariant. Counts below summarize the complete automata: reachable locations, transitions, and active clocks. The internal output-shape guarantee is proved in Rocq; the displayed graph and counts are measurements.

TABLE 3. Contract of the verified optimization and export pipeline. The explicit and symbolic outputs preserve the timed-word language of the TBA provided to the Rocq interface.

Aspect	Guaranteed result
Timed-word language	Preserved by every optimization round and by complete export under the development's existential initial-clock semantics.
Explicit guards	Conjunctions of event literals and clock comparisons; export may introduce clock-difference comparisons in entry conditions.
Location invariants	Conjunctions of non-strict clock upper bounds.
Symbolic output	Groups transitions with the same source, target, and reset set under Boolean labels; the symbolic automaton also preserves acceptance.
Input adapter	A third-party parser or conversion into the Rocq TBA representation requires a separate correspondence proof to join the verified boundary.

Proof. The translation theorem `MTL_to_TBA_correct_with` gives the first equivalence below. The other two are the preservation results of this paper, which do not depend on how C was produced. Let $C = \mathsf{compile_with}(A)$, $O_n = \mathsf{optimize}(n, C)$, and $D_n = \mathsf{export}(O_n)$. Then

$$\mathsf{msat}(w, 0, f) \leftrightarrow \mathsf{TBA_accepts}(C, w)$$
$$\mathsf{TBA_accepts}(C, w) \leftrightarrow \mathsf{TBA_accepts}(O_n, w)$$
$$\mathsf{TBA_accepts}(O_n, w) \leftrightarrow \mathsf{DTA_accepts}(D_n, w).$$

The repository theorem `MTL_to_exported_correct_with` performs this composition. This paper adds no proof of the MTL translation. □

The first hypothesis is exactly the backend contract of `MTL_to_TBA_correct_with`, required on the action-consistent propositional words (those in which the negated action atom is the complement of the action atom); `well_formed` requires nonnegative non-strict upper bounds and strictly positive bounds for the other primitive timed operators. No convergence premise is imposed on n: `optimize_accepts` proves preservation by induction for every finite round count, and `export_accepts` then preserves the optimized TBA's language. The implementation's 50-round stop is a practical driver policy, not part of this theorem.

The theorem's conclusion concerns the internal exported DTA record and the development's existential initial-clock semantics. It does not establish Spot's correctness unless the backend premise is separately discharged, nor does it prove correctness of the LBTT importer, XML printer, XML parser, or UPPAAL execution. A second mechanized theorem, `MTL_to_exported_correct0_with`, derives the corresponding zero-initialized result when `init_free`(export(optimize(n, compile_with(A)))) = `true`. The underlying lemma `init_free_accepts0` holds for every exported automaton on which the check succeeds; nothing guarantees that the check succeeds on an automaton whose acceptance relies on nonzero initial clock values.

The separately verified all-copy policy yields the parallel theorem `MTL_to_exported_all_correct_with`. For the same n, f, A, and w, under the same backend contract

and well_formed(f) hypothesis, it proves

$$\mathsf{msat}(w, 0, f) \leftrightarrow \mathsf{DTA_accepts}(\mathsf{export}_{\mathrm{all}}(B), w),$$
$$B = \mathsf{optimize}(n, \mathsf{compile_with}(A)).$$

Its conclusion has the same internal-DTA and existential-initial-clock boundary. The all-copy theorem does not strengthen the zero-initialization claim or verify XML serialization.

VII. TOOL ARCHITECTURE AND USE IN UPPAAL

A. IMPLEMENTATION AND PROOF BOUNDARY

The OCaml module `optim.ml` is extracted from Rocq and contains the verified optimization and export functions. The integrated executable also contains handwritten input, iteration, and output code. These adapters and the XML/DOT printers are outside the transformation proofs. A converter from an external automaton format must establish its own correspondence with the Rocq TBA type. A clean Rocq 9.0.1 build, independent `coqchk` run, and OCaml 5.4.0 rebuild are recorded in the accompanying artifact audit. The UPPAAL 5.0.0 queries were rerun in this revision and agree with the archived verdicts.

The project-wide axiom `LTL_TO_BUCHI_CORRECT` states the assumed correctness of the upstream LTL-to-Büchi backend. Convenience theorem variants use it when backend correctness is not an explicit argument. The theorems `optimize_accepts`, `split_accepts`, and `export_accepts` theorems quantify over arbitrary values of the formal interface and do not use that project axiom. The theorem `MTL_to_exported_correct_with` keeps the propositional-backend equivalence as an explicit premise. Its fresh assumptions audit reports standard classical Rocq library principles, including `ClassicalDedekindReals.sig_forall_dec`, but no project-specific backend axiom. The axiom `LTL_TO_BUCHI_CORRECT` appears in convenience theorems that discharge that premise using the project axiom. The artifact records the separate `Print Assumptions` results.

In the integrated workflow, the upstream producer supplies a TBA at the optimizer boundary after its own construction stages. The earlier article [1] specifies and proves the MTL-to-TBA translation; this paper takes that resulting TBA as an input and concerns only the subsequent optimizer and exporter. The driver iterates optimization until structural stability or a 50-round cap, then applies export. This cap is an implementation policy; language preservation holds for every finite round count.

The preservation theorems use existential initial clock values. UPPAAL starts clocks at zero. The verified `init_free` check bridges the two conventions for every exported automaton on which it succeeds; when it fails, which can happen for an external automaton whose acceptance relies on nonzero initial clock values, no zero-initialization conclusion is drawn.

B. INPUT CONTRACT FOR AN EXTERNAL TBA

The interface starts from a finite TBA value (L, ℓ_0, X, E, Acc) in the Rocq representation. Its clock set X is `Clockroot`, the timed subformulas of a formula `root` fixed by the development; an automaton with k clocks is represented by choosing any formula with at least k distinct timed subformulas, which then serve only as clock names. Its transitions carry event conditions, finite conjunctions of unary clock comparisons, and reset sets; the starting value has no location-invariant annotation. The invariant synthesis pass derives that annotation from outgoing guards. The theorem does not require a complete or deterministic transition relation, a particular number of clocks, or a connection to a temporal-logic translator. A third-party producer can therefore use the proved functions after converting its automaton to this representation and naming its clocks in this way.

The conversion is a semantic boundary: it must retain the alphabet and the meaning of each event condition, preserve which clocks are tested and reset, and map accepting locations without changing their marking. The paper's repository provides the Rocq type and extracted optimizer, but not a verified reader for arbitrary third-party formats. Any such reader belongs to the trusted interface until its correspondence with the Rocq value is proved. The same applies to interpreting the result in a target checker: export proves language preservation for the Rocq representation, while a separate bridge is needed to identify the checker's parsed model with that value. This boundary keeps the central optimization theorem independent of the producer while making the work needed to reuse it explicit.

C. INITIALIZATION INTERFACE

The preservation theorems use existential initial clock values, whereas UPPAAL initializes clocks to zero. The proved `MTL_to_exported_correct0_with` theorem gives the zero-initialized result for the formula-indexed workflow when the verified `init_free` check succeeds. All 29 completed benchmark outputs pass that check; the two timed-out configurations produce no output to check. For an external automaton the same check applies, and the zero-initialized conclusion is available only when it succeeds.

D. USING EXPORTED AUTOMATA IN UPPAAL

The language-preservation theorem concerns infinite timed words with one event at each position, strictly increasing timestamps, divergent time, and the development's clock initialization semantics. A UPPAAL composition must preserve the assumptions needed to interpret its traces in that model:

(C1) *Events and time.* The system must synchronize each observable event with the property automaton, prevent zero-delay consecutive events (for example, reset a system clock z on each emission and require $z > 0$ on the next), and admit time-divergent infinite executions. Reachability alone does not rule out a Zeno continuation.

(C2) *Initial clocks.* UPPAAL initializes clocks to zero, while the preservation theorems use existential initial values. The zero-initialized conclusion requires the `init_free` check of the preceding subsection to succeed on the exported automaton.

(C3) *Acceptance.* UPPAAL's ordinary reachability interface does not check Büchi acceptance. For safety, use a violation automaton with an accepting sink that is reached when a violation occurs and has a loop on every event. Liveness requires a Büchi emptiness checker that restricts attention to time-divergent runs; the artifact uses TChecker for this task.

(C4) *Time constraints in products.* Location invariants are part of the automaton semantics and can block time in a product, thereby removing system behaviors. An exported property automaton is not a passive observer unless its effect on the product is separately justified. Check deadlock or timelock conditions on the system as appropriate.

(C5) *Numbers.* The integrated tool accepts integer bounds in $[0, 2^{30}]$. The Rocq theorems are over real-valued clocks; the extracted implementation's arithmetic and the UPPAAL printer are covered only on the tool's restricted numeric domain.

The generated XML has a `Property` template for the automaton and an `Env` template that can emit events for standalone language exploration. In a system check, the system replaces `Env` and synchronizes its observable events with `Property`; the event `other` denotes any event outside the finite set named in the formula. Accepting locations are named A*n* in the output, but those names do not make UPPAAL check Büchi acceptance. For a safety requirement, the reachability query must use an automaton for its violations with a sink entered once the violation has occurred. For a liveness requirement, reachability of such a sink is not the right check and an emptiness procedure for time-divergent Büchi runs is needed.

For a safety property φ, let $\mathcal{A}_{\neg\varphi}$ be a violation automaton whose accepting sink has no invariant and loops on every event. Compose it with a system S that synchronizes the same event alphabet and can extend every queried prefix to a time-divergent execution. Under (C1)–(C3) and the appropriate initialization condition in (C2), the query `E<> P.A` detects a system behavior violating φ when the automaton recognizes exactly those violations. This query-level equivalence is an argument about the chosen system and automaton; it is not proved by the Rocq optimization or export theorem. The case-study models below report the model checks and diagnostic traces for the four packaged systems.

E. AVAILABILITY

This project contains proof sources, extracted implementation, benchmark tables and raw data, reproduction scripts, and UPPAAL case-study models, together with the LaTeX source and build instructions. The invariant, optimization, and export formalization files contain the generic pass results; their theorem names are listed in Table 1. The project repository is at https://github.com/efares1/rocq-tba-optimization-uppaal-export, and the repository of the translation article is at https://github.com/efares1/rocq-mtl-to-tba-translation. The revisions described here are the releases tagged `v1.1` in both repositories.

F. BUILD AND ARTIFACT LAYOUT

The verified functions are defined under `proof/` and extracted into `tool/src/optim.ml`. The project Makefile compiles the Rocq development and extracts the executable functions. Hand-written components include the parser, Spot invocation, LBTT reader, iteration driver, and XML/DOT printers; the script that removes unreachable extracted definitions and the Rocq/OCaml toolchains are also part of the implementation's trusted base. The repository includes the machine record, raw benchmark runs, TSV summaries, table generator, formulas, UPPAAL models and queries, and reproduction instructions in `README.md`.

Table 4 separates the verified transformations from the interfaces that connect them to the surrounding tools.

In the extracted code, structural equality decides formula and clock identity, and Boolean checks implement conditions whose soundness is proved in Rocq. Real comparisons are realized by OCaml floating-point operations. Input constants lie in $[0, 2^{30}]$; the inspected operations produce only such constants, their differences, or their negations. These values are exactly representable by the 53-bit significand of a double, and the printed constants fit UPPAAL's 32-bit integer domain. This arithmetic argument is based on inspection of the extracted implementation rather than a theorem about OCaml or the printer. Rational constants can be handled by a common time scaling only when every scaled bound remains an integer in $[0, 2^{30}]$; timing constants in any surrounding system model must be scaled by the same factor.

VIII. EVALUATION

A. QUESTIONS, EVIDENCE, AND SETUP

The evaluation asks four separate questions. First, which structures do the verified optimization passes remove or simplify? Second, what size and timing cost does conversion to conjunctive guards and invariants add? Third, do the packaged system checks behave as intended? Fourth, does optimization affect a matched downstream check? Internal output shape is a Rocq theorem; external checker verdicts and resource use are experimental evidence. The matched TChecker cases below show unchanged exploration on systems whose exported models are byte-identical before and after optimization; they do not establish a general cost improvement.

We retain the 31-attempt F/R/N benchmark as an upstream-generated group. The optimizer input is the automaton after reset completion, shown as Comp. in Table 5. Of the 31 attempted configurations, 29 completed this upstream stage and supplied an optimizer input; R7 and R8 timed out before the measured boundary. Formula rewriting, Spot configuration, cube expansion, relaxation, and reset completion

TABLE 4. Runtime stages and their proof boundary. The preservation theorems begin with a TBA in the Rocq representation; the Spot-based formula front end is one client of the verified optimizer and exporter.

Stage	Implementation	Guarantee or boundary
TBA input and adapter	Rocq TBA value or a parser/converter supplied by the caller	Pass results start at this representation; an external adapter is outside the theorem unless separately verified.
Spot-based front end	Spot call and hand-written LBTT reader in `mtl2tba`	Used by the integrated formula workflow; outside the TBA preservation theorems.
Optimization rounds	Extracted from `MTL_to_TBA_Optimizations.v`	`optimize_accepts`, for every TBA and every round count.
Iteration driver	Hand-written OCaml structural comparison; stops at equality or 50 rounds	The proof does not assume convergence or a particular stopping point.
UPPAAL export	Extracted from `MTL_to_TBA_Export.v`	`export_accepts` and conjunctive guard/invariant results.
Symbolic grouping	Extracted from `MTL_to_TBA_Symbolic.v`	`symbolic_ext_accepts`.
XML, DOT, and PDF output	Hand-written printers and Graphviz	Outside the Rocq language-preservation results.
Numeric realization	OCaml floating point with integer bounds in $[0, 2^{30}]$	Exactness follows from inspection of the extracted operations, not a mechanized theorem.

happen upstream; their effects are not credited to the optimizer. The runs use the tool's default front end, which before translation rewrites $\Box\Diamond_{\geq d}\varphi$ into $\Box\Diamond\varphi$ and $\Diamond\Box_{\geq d}\varphi$ into $\Diamond\Box\varphi$, and which replaces the Until of the clauses of upper-bounded hatted Until by a weak Until. The end-to-end theorem for this front end is `MTL_to_exported_correct0_recur_weak_with` (`MTL_to_TBA_Recur.v`); the optimizer theorems are unaffected, since they hold for every input TBA of the interface. Results come from the repository's five-run medians on Ubuntu 26.04/WSL 2, an Intel Core i7-8650U at 1.9 GHz with 11.7 GB memory, OCaml 5.4.0, Rocq 9.0.1, and Spot 2.16. The fresh run used five repetitions per completed configuration and a 300-second limit per attempt. Its run log, machine record, summary table, and generated automata are preserved under `evaluation/runs/2026-10-14-full-eval-v3/`.

Across the 29 completed inputs, the sum of reachable transition counts falls from 20,204 at the post-reset-completion boundary to 1,500 after optimization (an aggregate reduction of 92.58%). Nineteen cases have fewer transitions and 10 are unchanged. This aggregate is strongly skewed: R6 contributes 14,985 of the 18,704 removed transitions, so the percentage is not a typical per-instance reduction. Active clocks used in guards or resets sum from 77 to 74; only three cases (F2, F11, F13) reduce this count. The export stage has a different effect: explicit transitions sum from 1,500 to 3,209, an increase of 113.93%; six exports grow and 23 have the same count as their optimized input. Export-size change is reported separately from optimization and is not counted as an optimization regression. The family breakdown shows how concentrated these totals are. The six completed bounded-response R cases account for 18,441 of the 18,704 removed transitions (98.59%); the F and N groups remove 46 and 217, respectively. The R group's explicit export grows from 1,089 to 2,796 transitions, while the F group grows by two and the N group is unchanged. Thus the aggregate optimizer reduction is driven mainly by one bounded-response family, and the export expansion is a separate representation cost concentrated in that same family.

Part of this reduction only re-merges the cubes that Spot emits one by one. With symbolic grouping applied in both runs, the optimizer alone lowers the symbolic transition count (`-n 0` against the default) from 704 to 344 (51%) on the 27 configurations whose unoptimized export completes: from 364 to 216 on F and N, and from 340 to 128 on R1–R4. It is smaller in 17 and equal in 10 (supplement). The unoptimized export of R5 and R6 does not finish within 300 s, whereas the optimized pipeline completes them in 3.6 s and 39.0 s.

B. OPTIMIZATION EFFECTS ON THE GENERATED GROUP

In R6, Spot emits 64 states and 15,625 cube-split transitions. The recorded post-reset-completion optimizer input also has 64 locations and 15,625 transitions; the equality of these two edge counts is specific to R6, and the Comp. column is the authoritative optimizer boundary. Six optimization rounds reduce the transition relation to 640 edges and retain six clocks. Export then produces 1,837 explicit edges, 63 invariant-bearing locations, and 3,360 clock-difference guard occurrences; symbolic grouping represents the output with 448 transitions. The larger explicit edge count results from splitting disjunctive guards and adding entry conditions for invariant copies. It measures the cost of the target representation, not a reversal of the optimizer's reduction. In the fresh run, median optimizer and export times are 17.807 s and 10.666 s, respectively. The generated-group table separately reports the stage after reset completion so that upstream processing is not folded into these optimizer measurements.

The same distinction matters for the other generated inputs. For example, F11 starts from its actual four-location, seven-transition, two-clock TBA after reset completion; two rounds leave three locations and four transitions and remove one clock. Two locations have the same loop and the same exit on p to the accepting location, except that the exit of the first also carries $x \geq 2$; the second is entered only when $x \geq 2$ holds and x is not reset afterwards. They are merged because the guards are compared under that entry invariant, which equality of guards cannot do. N8 spends most of its end-to-end time in Spot, whereas its recorded optimizer and

TABLE 5. Fresh five-run post-processing measurements (12 October 2026) for 31 attempted upstream-generated configurations; 29 produced optimizer inputs. Comp. is the automaton after reset completion. L counts graph-reachable locations, E counts transitions from those locations, and C counts active clocks used in guards or resets; n_f is the formula-clock-key count. Opt. and Exp. give reachable $L/E/C$, changed optimization rounds, stage medians, invariant-bearing locations, and diagonal-guard occurrences. R7 and R8 time out upstream, before the optimizer boundary.

ID	Comp. $L/E/C$	n_f	Opt. $L/E/C$	Rnds	Opt. (s)	Exp. $L/E/C$	Inv.	Diag.	Exp. (s)
F1	2/5/1	1	2/5/1	1	< 0.01	2/5/1	1	0	< 0.01
F2	2/5/1	1	1/1/0	1	< 0.01	1/1/0	0	0	< 0.01
F3	2/5/1	1	2/5/1	1	< 0.01	2/5/1	0	0	< 0.01
F3b	1/1/0	1	1/1/0	0	< 0.01	1/1/0	0	0	< 0.01
F4	2/5/1	1	2/5/1	1	< 0.01	2/5/1	1	0	< 0.01
F5	7/38/1	1	7/25/1	1	< 0.01	7/25/1	0	0	0.01
F6	7/38/1	1	7/25/1	1	< 0.01	7/25/1	0	0	< 0.01
F7	2/4/1	1	2/4/1	1	< 0.01	2/4/1	1	0	< 0.01
F8	2/4/0	0	2/4/0	1	< 0.01	2/4/0	0	0	< 0.01
F9	4/9/2	2	4/9/2	1	< 0.01	4/9/2	2	0	< 0.01
F10	5/13/3	3	5/13/3	1	< 0.01	5/13/3	2	0	< 0.01
F11	4/7/2	2	3/4/1	2	< 0.01	3/4/1	1	0	< 0.01
F12	3/9/2	2	2/5/2	2	< 0.01	2/5/2	1	0	< 0.01
F13	3/6/2	2	2/3/1	2	< 0.01	2/3/1	1	0	< 0.01
F14	6/23/2	2	6/17/2	1	< 0.01	6/19/2	4	2	< 0.01
R1	2/5/1	1	2/5/1	1	< 0.01	2/5/1	1	0	< 0.01
R2	4/25/2	2	4/16/2	2	< 0.01	4/19/2	3	2	< 0.01
R3	8/125/3	3	8/44/3	3	0.01	8/67/3	7	30	0.01
R4	16/625/4	4	16/112/4	4	0.06	16/217/4	15	192	0.16
R5	32/3125/5	5	32/272/5	5	0.94	32/651/5	31	880	1.45
R6	64/15625/6	6	64/640/6	6	17.81	64/1837/6	63	3360	10.67
R7	–	–	–	–	–	–	–	–	–
R8	–	–	–	–	–	–	–	–	–
N1	3/5/1	1	3/5/1	1	< 0.01	3/5/1	1	0	< 0.01
N2	4/11/2	2	4/9/2	1	< 0.01	4/9/2	2	0	< 0.01
N3	5/21/3	3	5/16/3	1	< 0.01	5/16/3	3	0	< 0.01
N4	6/36/4	4	6/25/4	1	< 0.01	6/25/4	4	0	0.01
N5	7/57/5	5	7/36/5	1	0.01	7/36/5	5	0	0.02
N6	8/85/6	6	8/49/6	1	0.03	8/49/6	6	0	0.06
N7	9/121/7	7	9/64/7	1	0.10	9/64/7	7	0	0.23
N8	10/166/8	8	10/81/8	1	0.33	10/81/8	8	0	0.70

TABLE 6. Family aggregates derived from the completed rows of Table 5. Edge totals are measured at the post-reset-completion optimizer boundary, after optimization, and after explicit export.

Group	n	Input $\sum E$	Opt. $\sum E$	Reduction	Exp. $\sum E$	Cases $\downarrow$ / $=$ / $\uparrow$
F group	15	172	126	26.74%	128	7/8/0
bounded-response R group	6	19,530	1,089	94.42%	2,796	5/1/0
N group	8	502	285	43.23%	285	7/1/0

exporter stages take 0.33 s and 0.70 s. These cases show that the optimizer's cost depends on its input relation and that the upstream producer can dominate total tool time. They do not support a general runtime or size bound. With the tool's `-spot-level low` option (default unchanged), N8 takes 11.6 s instead of 98.3 s, and the total over the 29 completed configurations falls from 152 s to 58 s (one run against the five-run medians); optimized sizes are identical except for F5 and F6 (25 to 27 symbolic transitions). R6 is dominated by the optimizer and the export, and its faster run (37.9 s against 39.0 s) is not attributed to the level.

C. EXTERNAL OUTPUT-SIZE REFERENCE

CASAAL [3] implements the $\mathrm{MTL}_{0,\infty}$ construction of Li et al. [4] and prints automata with Boolean transition labels. It is not an interchangeable optimizer: it builds its automata by its own construction, so its sizes are a reference for the final output, not a baseline for the optimization passes. We ran it with mutually exclusive events, which matches the one-event-per-position words of our semantics, and compare reachable locations and symbolic transitions of our optimized output with CASAAL's output on the 27 configurations that both tools completed and that CASAAL encodes exactly. The counts coincide in 22 cases. For F5 and F6 our output is larger (7 locations and 25 symbolic transitions against 4 and 15); for F11 (3 and 4 against 4 and 7), F12 (2 and 5 against 3 and 9), and F14 (6 and 14 against 8 and 18) it is smaller. Clock counts coincide except for F11, where two optimization rounds reduce the two clocks of the post-reset-completion input to one while CASAAL uses two, and F8, where the difference comes from the upstream rewriting of $\Box\Diamond_{\geq 2}p$ into $\Box\Diamond p$ and is not credited to the optimizer. In R6 both outputs have 64 locations and 448 symbolic transitions, whereas the optimizer input has 15,625 explicit transitions. Whether the two tools accept the same timed words is a question about the translation, examined in the earlier article [1]; this comparison concerns sizes only.

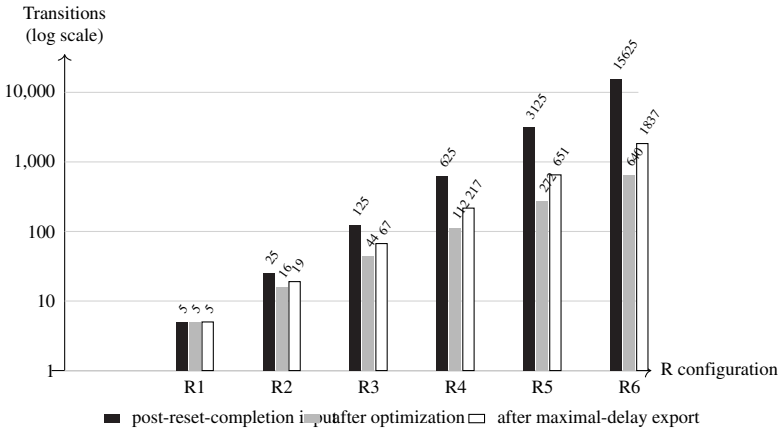


FIGURE 4. **Post-reset-completion input, optimized TBA, and maximal-delay explicit export for R1–R6. The logarithmic vertical axis makes the R6 concentration visible. These are the fresh five-run-median structural counts; Comp. is the optimizer boundary, regardless of whether an upstream Spot count happens to match it.**

D. MATCHED COMPARISON OF THE TWO EXPORT POLICIES

We reran R2–R6 once with each verified split policy while holding the optimized automaton fixed. For every formula, the optimized TBA DOT files were byte-identical across the two invocations. The table reports the explicit-edge counts, diagonal guard atoms, and single-run exporter time returned by the tool. Across these five inputs, all-copy export has no explicit-edge growth beyond the optimized relation and adds no diagonal guards. Maximal-delay export adds explicit edges in each case and introduces 2–3,360 diagonal guard atoms. This is a measured property of this benchmark family, not a general size theorem.

Both policies retain the same reachable location counts, active clock counts, and invariant-bearing locations on these runs. Peak RSS was measured for each full tool process, including Spot and optimization: 17.2/17.3 MB for R2 and 108.5/107.0 MB for R6 (maximal-delay/all-copy). These process-level measurements do not isolate export memory. The observed exporter times also have no consistent ordering: all-copy takes longer for R2/R3 and shorter for R4/R5, with similar times for R6. The R6 policy comparison uses the same 15,625-edge post-reset-completion input and six-round, 640-edge optimized TBA; the upstream Spot count is context, not the optimizer's pre-state.

E. TRANSLATOR-INDEPENDENT SYNTHETIC TBA SUITE

To exercise the interface without the MTL translator, we construct 16 small synthetic automata directly as Rocq-representation TBA records in `evaluation/synthetic_tba_bench.ml`. Their clocks are named by the six timed subformulas of a fixed formula, the conjunction of six hatted primitive formulas over three events; this formula is never translated and serves only as a supply of six clock names. Locations, event labels, unary guards, resets, and accepting locations are written by hand for each record. The suite is explicitly synthetic: it does not parse a third-party format and is not evidence of industrial-model performance. Its records cover redundant guards, contradictory guards, unsatisfiable labels, a source with no incoming edge, dead resets, synchronously reset clocks, duplicate transitions, equivalent locations, a disjunctive-invariant export, a stable no-change input, and targeted backward/forward propagation and pass-interaction cases.

The harness reports counts after every pass in every round and checks that its profiled pass sequence is structurally equal to the extracted `optimize_step` on each round. It also times the unprofiled iterative optimizer and export as means over 200 repeated in-memory calls per model. Standalone process RSS ranges from 6,184 to 6,484 KiB. These short timings and small, purpose-built records are microbenchmark evidence only; the raw pass table and per-model RSS file are supplied for reproduction. Per-pass times are recorded but are too small to support fine-grained speed rankings.

The targeted examples exercise both bound-propagation directions. Backward synthesis tightens an incoming $x \leq 5$ guard to $x \leq 2$. Forward propagation adds an entry lower bound $x \geq 2$ to outgoing edges. In the forward-cascade input, an alternate $x \leq 10$ branch prevents backward synthesis from creating the contradiction; forward propagation makes the $x \leq 1$ branch contradictory, and the following pruning stages remove its unentered chain. Omitting forward propagation retains all seven input edges, while the full optimizer leaves four. A separate backward cascade isolates that pass. Table 9 shows input/output counts and leave-one-operation-out effects.

The normalization/reset example removes a tautological $x \geq 0$ guard before dead-reset analysis, allowing its reset and active clock to be removed. The synchronous-clock example first merges the clocks, then a later round detects the conflicting $x \geq 5 \wedge x \leq 3$ guard and prunes the branch. Other records measure unsatisfiable-label elimination, no-incoming-source pruning, transition subsumption, and equivalent-location merging. The no-incoming operation

TABLE 7. Matched maximal-delay and all-copy exports for R2–R6. The two policies receive byte-identical optimized TBA graphs. Export time is one observed run per row and policy; it is descriptive and should not be read as a stable speed ranking.

ID	Optimized E	Max-delay E	All-copy E	Max-delay diagonal atoms	All-copy diagonal atoms	Max-delay time (s)	All-copy time (s)
R2	16	19	16	2	0	0.002	0.005
R3	44	67	44	30	0	0.014	0.024
R4	112	217	112	192	0	0.195	0.190
R5	272	651	272	880	0	1.319	1.052
R6	640	1,837	640	3,360	0	9.689	9.614

TABLE 8. Sixteen translator-independent synthetic TBA records. Count tuples are declared locations / transitions / distinct clocks / reset occurrences / unary guard atoms; the export tuple adds diagonal guard atoms. The *I* field gives nonempty invariant locations:bound atoms. Optimization and export times are per-call means over 200 repeated in-memory runs; RSS is the peak for a separate single-model process. Full per-pass and reachability data are supplied in the artifact.

Model	Input	Optimized	Rounds	Opt. (ms)	RSS (KiB)	Export	Exp. (ms)
stable	1/1/0/0/0	1/1/0/0/0; $I = 0:0$	0	0.008	6364	2/1/0/0/0/0; $I = 1:0$	0.089
redundant_guard	2/2/1/1/3	2/2/1/1/2; $I = 2:2$	1	0.133	6460	4/2/1/1/2/0; $I = 2:2$	0.197
contradictory_guard	2/2/1/0/2	2/1/0/0/0; $I = 0:0$	1	0.042	6428	4/1/0/0/0/0; $I = 1:0$	0.073
unsatisfiable_label	2/2/0/0/0	2/1/0/0/0; $I = 0:0$	1	0.026	6352	4/1/0/0/0/0; $I = 1:0$	0.143
isolated_source	3/2/0/0/0	3/1/0/0/0; $I = 0:0$	1	0.027	6340	6/1/0/0/0/0; $I = 1:0$	0.264
dead_reset	2/2/1/2/0	2/2/0/0/0; $I = 0:0$	1	0.052	6468	4/2/0/0/0/0; $I = 2:0$	0.150
normalization_dead_reset	2/2/1/1/1	2/2/0/0/0; $I = 0:0$	1	0.110	6336	4/2/0/0/0/0; $I = 2:0$	0.491
synchronous_clock_merge	2/3/2/2/2	2/3/1/1/2; $I = 1:1$	2	0.673	6300	4/3/1/1/2/0; $I = 2:1$	1.279
duplicate_transition	2/3/1/2/3	2/2/1/1/2; $I = 2:2$	1	0.267	6364	4/2/1/1/2/0; $I = 2:2$	0.478
equivalent_locations	3/4/0/0/0	3/2/0/0/0; $I = 0:0$	2	0.104	6404	6/2/0/0/0/0; $I = 2:0$	0.220
disjunctive_invariant_export	3/3/2/0/2	3/3/2/0/2; $I = 0:0$	0	0.095	6488	9/6/2/0/12/6; $I = 3:2$	1.154
backward_tightening	2/2/1/0/2	2/2/1/0/2; $I = 2:2$	1	0.093	6360	4/2/1/0/2/0; $I = 2:2$	0.204
forward_lower_bound	3/3/1/0/2	3/3/1/0/4; $I = 2:2$	1	0.141	6460	6/3/1/0/4/0; $I = 3:2$	0.654
forward_contradiction_cascade	6/7/1/0/3	6/4/1/0/4; $I = 1:1$	2	0.437	6428	12/4/1/0/4/0; $I = 3:1$	1.082
backward_contradiction_cascade	5/5/1/0/4	5/1/0/0/0; $I = 0:0$	2	0.201	6320	10/1/0/0/0/0; $I = 1:0$	0.180
clock_merge_contradiction_cascade	5/5/2/6/4	5/3/1/2/1; $I = 1:1$	2	0.622	6484	10/2/1/2/1/0; $I = 2:1$	0.598

is a local predecessor test, not full graph reachability. These interactions explain the iterative design; preservation for each finite iteration count follows from the proved pass composition.

Table 10 gives the measured stage interaction for the largest synchronously reset synthetic input. Unsatisfiable-label removal first reduces the relation from seven edges to five. Dead-reset elimination then lowers the active-clock count from six to four. The synchronous-clock pass changes the synthesized invariant, and the next round's dead-reset pass removes the remaining redundant resets, leaving one active clock. This is a measurement of the extracted implementation; the language-preservation claim follows separately from the proved pass composition.

The synthetic disjunctive-invariant case is unchanged by optimization, then grows from three locations and three transitions to nine locations and six explicit transitions after export. The output has conjunctive invariants and six diagonal guard atoms that select maximal-delay copies. This is a measured size cost of export; the output-shape guarantee itself follows from `export_guards_conjunctive` and `export_invariants_conjunctive`.

F. PASS ABLATION AND PARAMETERIZED SYNTHETIC FAMILIES

To measure pass interactions, we repeated the same composite pipeline while omitting one operation at a time from sixteen fixed synthetic input records. Each modified composite was iterated to structural stability (at most 50 changed rounds); this is a measurement driver, not a separate Rocq theorem about the ablated variants. The full variant was checked for structural equality with the extracted optimizer on each input. Table 11 reports aggregate reachable locations, total transition records, active clocks, changed rounds, and median repeated-call time. The raw TSV gives each record and its counts under every pass.

The aggregate ablation shows effects that selected records were designed to expose. Relative to the full variant (33 transitions and nine active clocks summed across the suite), omitting backward synthesis leaves 34 transitions and ten clocks; omitting forward propagation leaves 36 transitions; and omitting contradiction removal leaves 43 transitions. Omitting pruning of sources without incoming edges leaves 39 transitions. Normalization and dead-reset omission raise the aggregate active-clock count to 12 and 13, respectively. Omitting synchronous-clock merging leaves 35 edges and 11 clocks; omitting transition merging leaves 35 edges, while omitting equivalent-location merging leaves one more reachable location. These are descriptive differences on a deliberately selected synthetic suite, not estimates of typical pass effects. Per-call medians are too small for fine-grained pass runtime rankings; process RSS includes runtime overhead.

We also constructed five size-parameterized families directly in the formal TBA representation, without a third-party parser. Table 12 reports every tested size. In the duplicate-edge family, inputs grow from two to 33 edges while optimization reduces each nontrivial instance to two. The unreachable-cascade sweep includes sizes 1–50, 51, and

TABLE 9. Measured direct TBA examples targeting propagation and pass interactions. Transition and clock counts are total records and active clocks. Leave-one-operation-out variants are experimental composites, not separately verified transformations.

Case	Input E/C	Full E/C	Rounds	Measured target-pass effect when omitted
Backward bound tightening	2/1	2/1	1	Incoming guard tightens from $x \leq 5$ to $x \leq 2$; omitting backward synthesis leaves $x \leq 5$.
Forward lower-bound propagation	3/1	3/1	1	Entry fact $x \geq 2$ is added to the outgoing q edge; edge count is unchanged.
Forward contradiction cascade	7/1	4/1	2	Omitting forward propagation or contradiction removal retains 7 edges; the full run removes the $x \leq 1$ branch and then prunes its unentered chain.
Backward contradiction cascade	5/1	1/0	2	Omitting backward synthesis leaves 2 edges/1 clock; omitting contradiction removal leaves 5/1.
Normalization and dead reset	2/1	2/0	1	Normalizing tautological $x \geq 0$ enables reset removal; omitting either pass leaves 1 active clock.
Synchronous-clock contradiction	5/2	3/1	2	Merging the synchronously reset clocks exposes $x \geq 5 \wedge x \leq 3$; omitting clock merge or contradiction removal leaves 5 edges/2 clocks.

TABLE 10. Measured trace for the six-clock synchronous-reset synthetic input. Counts are declared / reachable locations, transitions, active clocks, reset occurrences, guard atoms, invariant locations / atoms, and diagonal guard atoms. The export row is the internal explicit DTA.

Stage	L/R	E	C	Resets	Guard atoms	Inv. L/A	Diag.
Input	2/2	7	6	6	6	0/0	0
R1: unsatisfiable-label removal	2/2	5	6	6	4	0/0	0
R1: dead-reset elimination	2/2	5	4	4	4	0/0	0
R1: synchronous-clock pass	2/2	5	4	4	4	1/1	0
R2: dead-reset elimination	2/2	5	1	1	4	1/1	0
Optimized fixed point	2/2	5	1	1	4	1/1	0
Explicit export	4/2	5	1	1	4	2/1	0

TABLE 11. Iterated leave-one-operation-out ablations on sixteen fixed synthetic TBA records. Reachable-location, transition, and active-clock counts are summed across the same inputs; rounds and runtime are medians across inputs. Runtime is the median of each model's mean over 200 in-memory runs. Each variant repeats its retained composite to structural stability. The full variant is checked against the extracted optimizer; the modified ablation driver is not a separate Rocq theorem.

Variant	$\sum L_{reach}$	$\sum E$	$\sum C$	med. rounds	med. time (ms)	cases with E above full
Full optimizer	30	33	9	1	0.102	0
Omit bound synthesis/backward propagation	31	34	10	1	0.104	1
Omit forward propagation	33	36	9	1	0.085	1
Omit contradictory-edge removal	40	43	11	1	0.102	4
Omit unsatisfiable-label removal	31	34	9	1	0.105	1
Omit no-incoming-source pruning	30	39	10	1	0.098	4
Omit guard normalization	30	33	12	1	0.107	0
Omit dead-reset removal	30	33	13	1	0.101	0
Omit synchronous-clock merging	32	35	11	1	0.091	1
Omit transition merging	30	35	9	1	0.084	2
Omit equivalent-location merging	31	35	9	1	0.068	1

64; rows at the 50-round driver cap are marked and report capped, not necessarily stable, outputs. A separate probe confirms that size 51 still changes on round 51 (two edges remain after round 50), while size 64 still changes on rounds 51 and 52. The size-50 chain has one edge after round 50 and is unchanged by round 51, but the driver conservatively warns because it did not perform that stability check. Synchronously reset families with two to six clocks reduce to one active clock in two rounds. The disjunctive-invariant export grows from 10 edges and 10 diagonal guard occurrences at $k = 2$ to 68 edges and 204 diagonal occurrences at $k = 6$. These counts expose representation costs in a deliberately constructed stress family; they are not industrial-model measurements or a complexity bound. For every finite number of rounds, including a capped result, `optimize_accepts` preserves language. Reaching the iteration cap does not justify calling that result a fixed point. Per-size intermediate records and the exact cap probe are preserved in the dated run artifacts.

G. UPPAAL AND TIMED-LIVENESS CHECKS

The four UPPAAL configurations are two parameterized system families: a sporadic emitter with minimum separation two (satisfying) or one (violating), and a request/response controller with response bound four (satisfying) or six (violating). The property query reaches a designated violation sink; it does not ask UPPAAL to check Büchi acceptance. The system-only `A[] not deadlock` query is satisfied for all four models; that result does not establish time divergence. The packaged traces show the violating event pairs. The result files contain verdicts and traces, not before/after optimization runtimes, memory, zone, or state counts, so they demonstrate use of the exported models rather than a quantified model-checking speedup.

The eight queries (four reachability, four deadlock) were

TABLE 12. Complete parameter sweeps for five translator-independent synthetic TBA families. Tuples report declared locations / graph-reachable locations / transitions / active clocks. The optimizer count is the number of structurally changed rounds; optimization time is the mean over 200 in-memory calls. Export diagonals are explicit clock-difference guard occurrences.

Family	k	Input $L/R/E/C$	Opt. $L/R/E/C$	Rnds.	Export $L/R/E/C$	Opt. (ms)	Diag.
Duplicate edges	1	2/2/2/0	2/2/2/0	0	4/2/2/0	0.017	0
Duplicate edges	2	2/2/3/0	2/2/2/0	1	4/2/2/0	0.030	0
Duplicate edges	4	2/2/5/0	2/2/2/0	1	4/2/2/0	0.046	0
Duplicate edges	8	2/2/9/0	2/2/2/0	1	4/2/2/0	0.057	0
Duplicate edges	16	2/2/17/0	2/2/2/0	1	4/2/2/0	0.073	0
Duplicate edges	32	2/2/33/0	2/2/2/0	1	4/2/2/0	0.216	0
Unreachable cascade	1	3/1/2/0	3/1/1/0	1	6/1/1/0	0.027	0
Unreachable cascade	2	4/1/3/0	4/1/1/0	2	8/1/1/0	0.075	0
Unreachable cascade	4	6/1/5/0	6/1/1/0	4	12/1/1/0	0.253	0
Unreachable cascade	8	10/1/9/0	10/1/1/0	8	20/1/1/0	1.752	0
Unreachable cascade	16	18/1/17/0	18/1/1/0	16	36/1/1/0	9.324	0
Unreachable cascade	32	34/1/33/0	34/1/1/0	32	68/1/1/0	47.381	0
Unreachable cascade	50	52/1/51/0	52/1/1/0	50	104/1/1/0	227.747	0
Unreachable cascade	51	53/1/52/0	53/1/2/0	50	106/1/1/0	239.316	0
Unreachable cascade	64	66/1/65/0	66/1/15/0	50	132/1/1/0	531.813	0
Redundant bounds	1	2/2/2/1	2/2/2/1	0	4/2/2/1	0.017	0
Redundant bounds	2	2/2/2/1	2/2/2/1	1	4/2/2/1	0.025	0
Redundant bounds	4	2/2/2/1	2/2/2/1	1	4/2/2/1	0.027	0
Redundant bounds	8	2/2/2/1	2/2/2/1	1	4/2/2/1	0.036	0
Redundant bounds	16	2/2/2/1	2/2/2/1	1	4/2/2/1	0.034	0
Synchronous resets	2	2/2/3/2	2/2/3/1	2	4/2/3/1	0.230	0
Synchronous resets	3	2/2/4/3	2/2/4/1	2	4/2/4/1	0.316	0
Synchronous resets	4	2/2/5/4	2/2/4/1	2	4/2/4/1	0.318	0
Synchronous resets	6	2/2/7/6	2/2/5/1	2	4/2/5/1	0.618	0
Disjunctive invariant	2	2/2/3/2	2/2/3/2	0	6/3/10/2	0.030	10
Disjunctive invariant	3	2/2/4/3	2/2/4/3	0	8/4/30/3	0.047	60
Disjunctive invariant	4	2/2/5/4	2/2/4/3	1	8/4/30/3	0.117	60
Disjunctive invariant	6	2/2/7/6	2/2/5/4	1	10/5/68/4	0.213	204

TABLE 13. UPPAAL safety-query results for the two packaged system families. The queries were rerun with the UPPAAL 5.0.0 verifier (`verifyta`) on the packaged models; the verdicts are identical to the earlier archived ones.

Family	Configuration	Query	Verdict
Sporadic emitter	Minimum separation 2	`E<> P.A4`	Not satisfied (expected)
Sporadic emitter	Minimum separation 1	`E<> P.A4`	Satisfied (violation trace)
Request/response	Response bound 4	`E<> P.A4`	Not satisfied (expected)
Request/response	Response bound 6	`E<> P.A4`	Satisfied (violation trace)

rerun with UPPAAL 5.0.0 (revision 714BA9DB36F49691, June 2023) on the packaged models; all eight verdicts coincide with the archived ones.

For liveness, we reran the separate TChecker timed Büchi emptiness case and added a matched before/after comparison. Before uses zero optimizer rounds; after requests the tool's 50-round maximum. Both formula-generated outputs satisfy `init_free`=1, so their zero clock initialization matches the TChecker product's zero-start convention for these cases. The system template was extracted from the archived UPPAAL model by the unverified `tck_normalize_system.py` adapter. The same system, property automaton, event synchronization, strict timestamp process, time-divergence process, and TChecker settings were used on both sides.

In both pairs the after run reported one changed optimizer round, but its exported XML hash matched the before XML, and the generated TChecker products were byte-identical. The checker explored the same states and transitions and returned the same verdict. These measurements show no downstream structural or exploration change for these examples; they do not demonstrate a speedup or a regression. TChecker checks nonemptiness of a timed-Büchi product over strictly increasing, time-divergent words. This remains distinct from UPPAAL's ordinary reachability and deadlock queries. Checker verdicts are experiments, not Rocq proofs. XML printing, the adapter, and format correspondence remain outside the mechanized interface; Rocq proves shape properties only for the internal DTA.

H. LIMITATIONS

The generated benchmark uses one upstream toolchain and the independent suite uses deliberately small synthetic inputs. The independent synthetic families demonstrate direct applicability to selected input shapes, but do not establish broad empirical support for arbitrary automata. No verified third-party parser or broad external TBA corpus is included. Export can grow both the state and transition representations. The matched TChecker study is limited to two response-system cases whose exported products were identical, and therefore gives no evidence of a performance benefit. Performance conclusions are limited to these measured inputs. The preservation theorems are universal only over the formal Rocq TBA interface and its timed-word semantics; that scope does not extend through an unverified adapter to arbitrary file formats.

TABLE 14. Matched TChecker timed-Büchi emptiness before and after the optimizer. State counts are stored/visited; runtime is the checker's reported time and peak RSS is the whole TChecker process. The exported XML and product automata are byte-identical within each pair.

System	`CYCLE` before/after	Stored/visited states before/after	Visited transitions	Runtime before/after (ms)	Peak RSS before/after (KiB)
Responds within 4 units	false / false	77/77 / 77/77	127/127	0.542 / 0.844	8,136 / 8,144
May remain busy forever	true / true	15/7 / 15/7	19/19	0.138 / 0.105	8,120 / 8,016

IX. RELATED WORK

Prior work already contains several of the algorithmic ideas used here. Daws and Yovine combine active-clock detection with detection of clocks that remain equal, and evaluate the resulting state-space reductions [5]. Their work is direct prior art for clock reduction; this paper does not claim that removing inactive clocks or merging equal clocks is a new algorithm. The Rocq development instead proves its specific dead-reset and synchronous-reset transformations as timed-word language equivalences and composes them with the other TBA passes. Herrera, Westphal, and Podelski extend equal-clock reduction to quasi-equal clocks [14].

Subsequent work makes the distinction between a local reduction pass and a global clock-minimization objective especially clear. Guha, Narayan, and Arun-Kumar construct a timed-bisimilar automaton with the least possible number of clocks, with a doubly exponential worst-case construction [6]. Yalcinkaya and Aydin Gol give a position-based clock-reduction method that preserves design parameters and prove timed bisimilarity [7]. These results are stronger objectives than the particular synchronous-reset merge of this paper: it does not minimize clock count, but proves language preservation for this pass together with the interacting TBA transformations. Neither cited article describes a proof-assistant mechanization.

Badban, Leue, and Smaus's CIPM algorithm derives control-location invariants using existing invariants and constraints on incoming transitions, and prunes idle transitions [8]. Their target is invariant generation for predicate abstraction, not a generic Büchi-language preservation contract. They explicitly note that UPPAAL does not admit disjunctive location invariants, approximate such joins by `true`, and mention location splitting as a possible direction; the paper does not give the split construction formalized here [8]. This observation is direct prior context for the representation problem, so the present work does not claim that splitting locations is a new idea.

De Wulf, Doyen, and Raskin study a compositional encoding of the almost-ASAP (AASAP) semantics for Elastic controllers [9]. Their construction uses location copies to represent phases of the controller encoding. The input controllers have no location invariants and use closed guards; the timed-automaton formalism allows multirectangular (disjunctive rectangular) predicates. The construction uses `PreIn`, `In`, `Out`, and `PostOut` phase copies, coordinated by internal labels such as τ, in_ℓ, and out_ℓ. These copies do not correspond one-to-one with invariant disjuncts: an `Out` location may retain a multirectangular invariant. Their correctness theorem relates the specialized AASAP controller-safety semantics to the synchronized encoding. This is relevant prior work on timed-automaton representation and disjunctive predicates, but it does not prove generic timed-word Büchi-language preservation or conjunctive output shape for arbitrary TBAs. This paper does not claim that location copying is new; it proves two split policies for its formal DTA interface and introduces no silent labels.

Bodeveix and Filali's note [10] informally proposes a one-copy-per-disjunct split for two disjuncts in UPPAAL and announces a PVS proof still ongoing; we make no priority claim for location splitting. Our searches for verified or mechanized timed-automata clock reduction and invariant transformations found no mechanized composition of such passes.

Mechanized timed-automata work addresses adjacent guarantees. Wimmer and Lammich refine timed-automata reachability checking to an executable Isabelle/HOL model checker, MUNTA [11], [12]. Wimmer, Herbreteau, and van de Pol verify an Isabelle certifier for emptiness certificates of timed Büchi automata, including certificates extracted from TChecker and Imitator runs [13]. These works verify checking or result certification; they do not establish the optimization and representation-transformation contracts proved here.

The defensible novelty claim is primarily mechanization and composition: individual simplification and invariant-generation ideas have substantial prior art, while this development proves their interaction and finite-round language preservation for the formal TBA interface, proves disjunctive invariant output shape, and connects the results to extracted code. The 29 completed upstream inputs and directly constructed synthetic suite support benchmark-specific practical claims only. They do not establish broad third-party format coverage or universal efficiency.

X. CONCLUSION

We presented a Rocq-verified optimization and export pipeline for timed Büchi automata. It synthesizes and propagates invariants, simplifies timing constraints, removes infeasible edges and dead resets, merges equivalent clocks and structure, and eliminates disjunctive invariants. Optimization preserves language for every TBA of the formula-indexed clock interface and every finite number of rounds; export preserves language and satisfies verified output-shape properties.

The composition theorem connects the optimization/export result to the translation of [1] under its explicit backend-language contract. The extracted implementation is used by `mtl2tba`. Measurements show large transition reductions for some inputs and costs from conjunctive export, includ-

TABLE 15. Scope comparison with representative prior work. "Not reported" means not stated in the cited source, not that no related result exists.

Prior work	Method and setting	Stated guarantee and mechanization	Difference from this paper
Daws–Yovine [5]	Active-clock reduction and equal-clock merging in timed automata	Algorithmic reduction with case studies; proof-assistant mechanization not reported	Prior clock reduction; no generic TBA pipeline theorem
Guha–Narayan–Arun-Kumar [6]	Minimum-clock timed-bisimilar automaton	Least clock count; doubly exponential construction; mechanization not reported	Stronger minimization objective, not verified passes/export
Yalcinkaya–Aydin Gol [7]	Position-based reduction preserving design parameters	Timed bisimilarity and measured clock reduction; mechanization not reported	Direct clock-reduction prior art; this work verifies a composed pipeline
Badban–Leue–Smaus [8]	CIPM invariant generation and idle-transition pruning; split discussed as a direction	Predicate-abstraction method; no split theorem or mechanization reported	Relevant disjunction problem, not generic TBA split preservation
De Wulf–Doyen–Raskin [9]	AASAP phase copies, watcher product, multirectangular predicates	Safety equivalence for specialized encoding; mechanization not reported	Phase copies, not one-copy-per-disjunct TBA normalization
Bodeveix–Filali [10]	Per-disjunct split of UPPAAL disjunctive invariants	Informal; PVS proof announced, ongoing	Prior splitting; no priority claim
Wimmer–Lammich / MUNTA [11], [12]	Timed-automata reachability checking	Verified checker and executable refinement in Isabelle/HOL	Proves analysis, not TBA optimization/export
Wimmer–Herbreteau–van de Pol [13]	Timed Büchi emptiness certificates	Sound, complete certificate checking in Isabelle/HOL	Verifies checking, not automaton transformation
This paper	Iterated optimizer, two split policies, conjunctive explicit/symbolic export	Rocq language preservation for each finite round/export and target-shape theorems	Every TBA of the formula-indexed clock interface; excludes adapters and XML

ing an R6 increase from 640 optimized transitions to 1,837 explicit transitions. Four UPPAAL configurations provide reachability/deadlock evidence, and a matched TChecker timed-Büchi comparison gives the same verdicts and exploration before and after optimization on two response-system cases whose exported XML is identical. This negative result supports no speedup claim. The UPPAAL queries were rerun with the installed verifier and give the archived verdicts.

Future work includes evaluating the passes on independent TBA corpora, improving symbolic handling of Boolean transition labels before Spot expansion, extending clock merging beyond synchronous reset, and verifying additional input/output adapters so third-party formats can use the pipeline with smaller trusted interfaces. We also leave to later work matched UPPAAL and TChecker speed-up runs, a converter from CASAAL's output to the automaton type of the optimizer, timing repeated over several runs, and faster handling of the R family, for example with symbolic transitions and generalized Büchi acceptance on transitions.

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